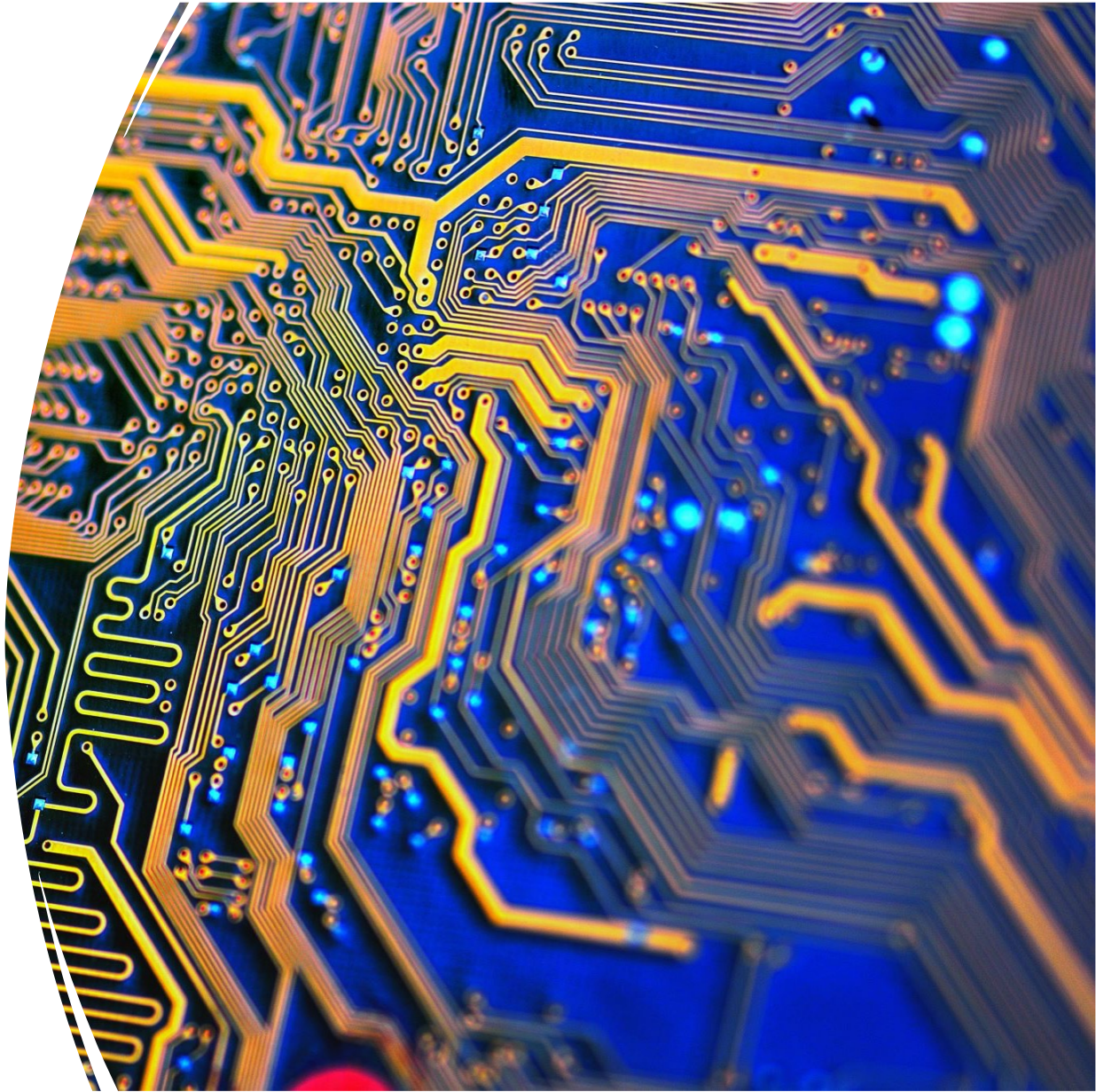


Unit 4

SOCIAL INTERACTIONS



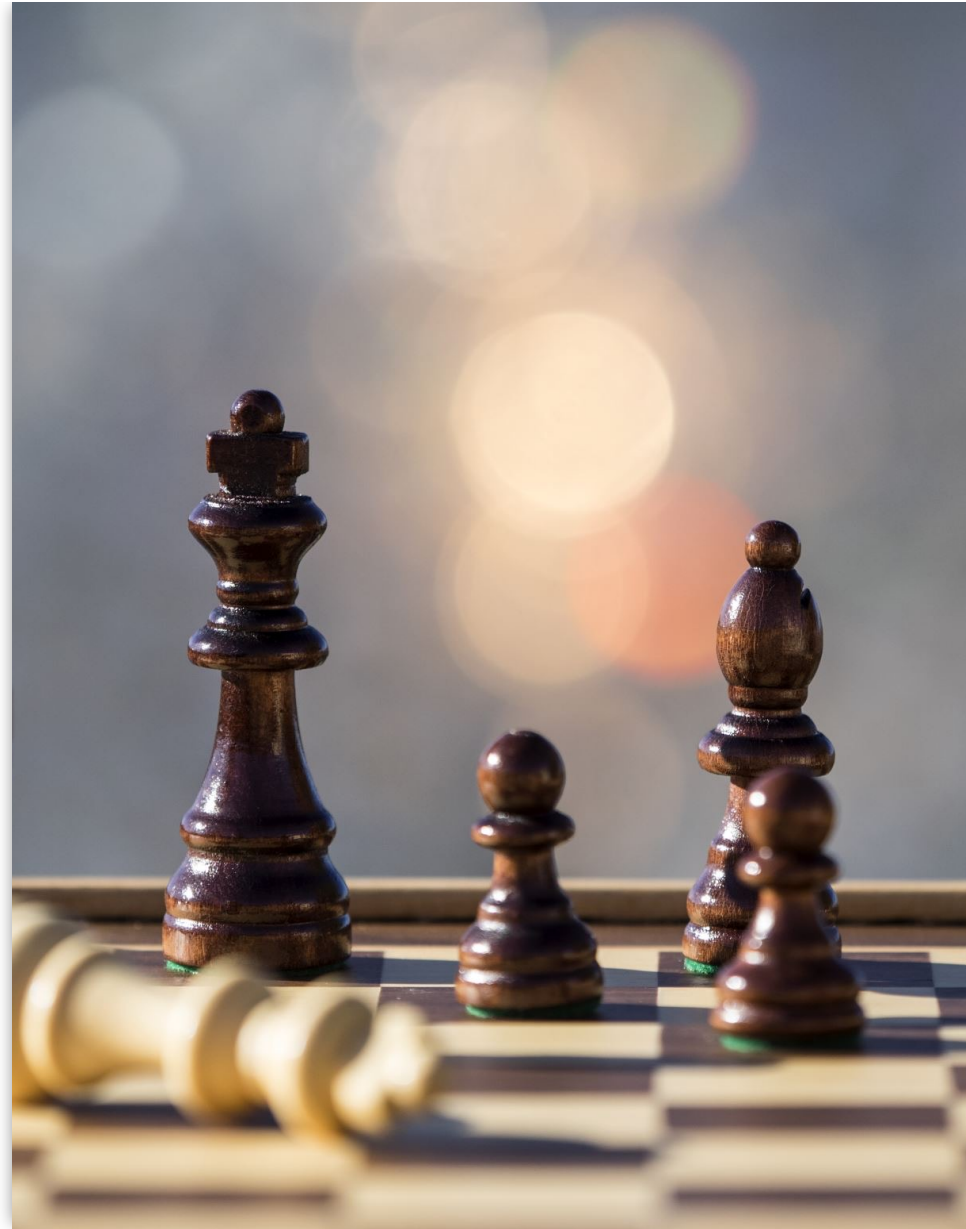
Objectives of Unit 4

- Social Dilemmas
- Game theory concepts
- Introduction of games
 - Invisible hand game
 - Prisoner's Dilemma: [Two players](#)
 - Pareto criteria
 - Altruism
 - Ultimatum game
 - Coordination Game (Programming game)



NB concepts for this Unit:

- [Game](#)
- [Best response](#)
- [Dominant strategy equilibrium](#)
- [Social dilemma](#)
- [Altruism](#)
- [Reciprocity](#)
- [Nash equilibrium](#)
- [Prisoners' dilemma](#)
- [Pareto efficiency](#)



The context for this unit

- Previous models of choice (Unit 3) did not depend on *others' decisions*
- Individuals motivated by self-interest can produce outcomes that are beneficial for society e.g. entrepreneurship, innovation (Unit 1)
- However, self-interest can also be harmful to society (create social dilemmas):
 - Why do these problems arise?
 - What can we do about it?



Climate negotiations

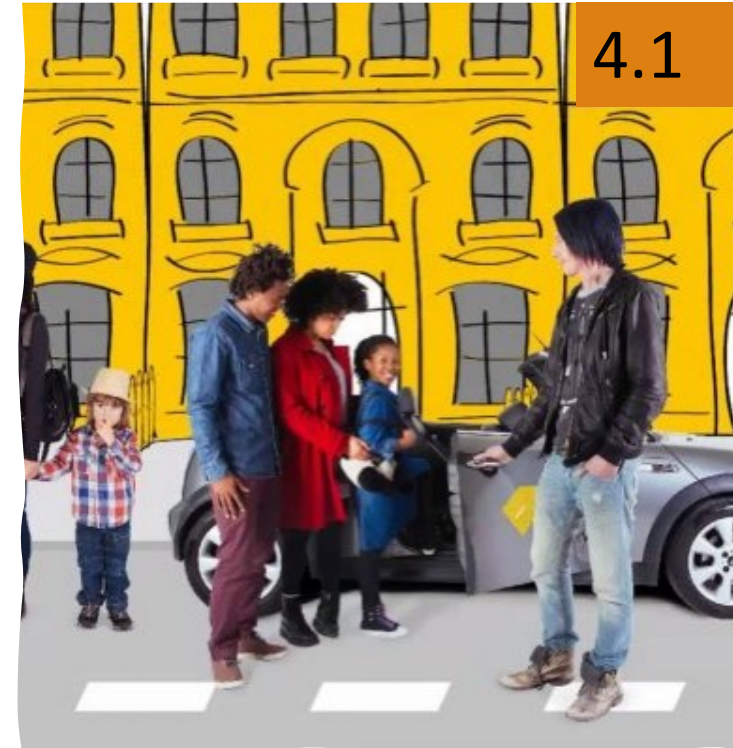
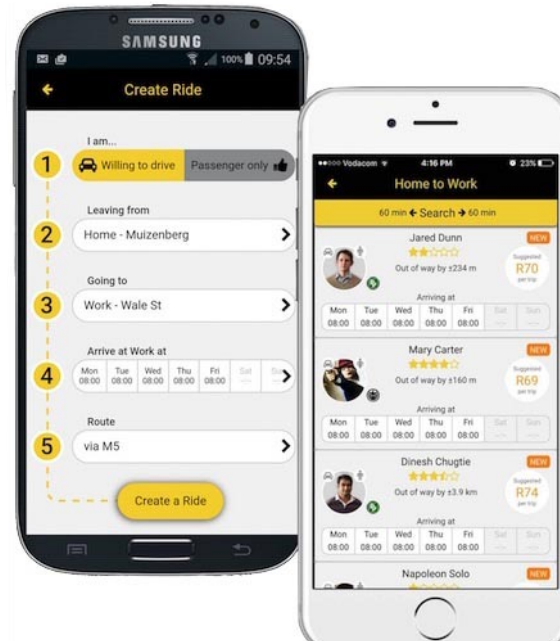
- Climate change presents serious global risks
 - Must cut greenhouse gas emissions
 - Reduce consumption of energy-intensive goods
 - Use different technologies to prevent environmental degradation
 - Improve efficiency in current technologies
- Cannot happen under “business as usual”
- National governments disagree on which policies to adopt
- 2015 Paris agreement, but it is not binding
- This leads to a **social dilemma** where people do not take adequate account of effect of their actions on others

Social dilemmas

- **Social dilemma** = a situation in which actions taken independently by self-interested individuals result in a socially suboptimal outcome e.g. traffic jams, climate change.
 - Forms of social dilemmas:
 - Tragedy of the commons
 - Free riding
 - How does society address social dilemmas?
 - Example: Institutions
- For more discussion on social dilemmas refer to video 1 :



Solutions to social dilemmas: Congestion



Solutions to social dilemmas



Around 3400 people die in traffic clashes every day, leading to 1.25 mill deaths globally each year

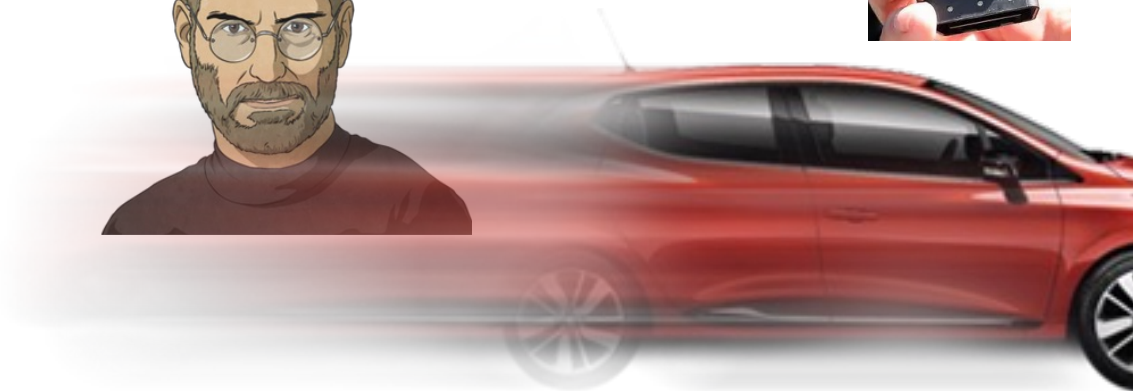


Switzerland holds the [world record for a speeding ticket](#). It was handed to a Swedish motorist in 2010 who was caught driving at 290km/h. He was fined 3,600 Swiss francs per day for 300 days – around 1,080,000 Swiss francs (\$1,091,340) in total.



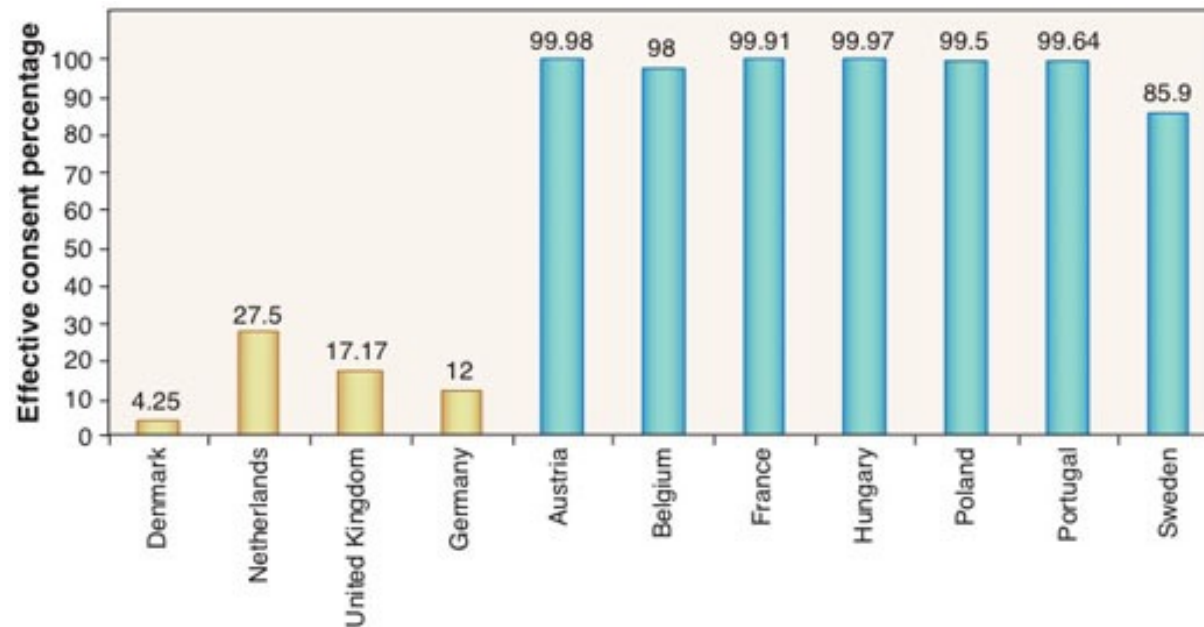
Similarly, Finland applies progressive punishment, speeding tickets are linked to your salary

Research has shown wealthier people tend to drive more recklessly; Steve Jobs was known to park in handicapped spots and drive around without license plates.

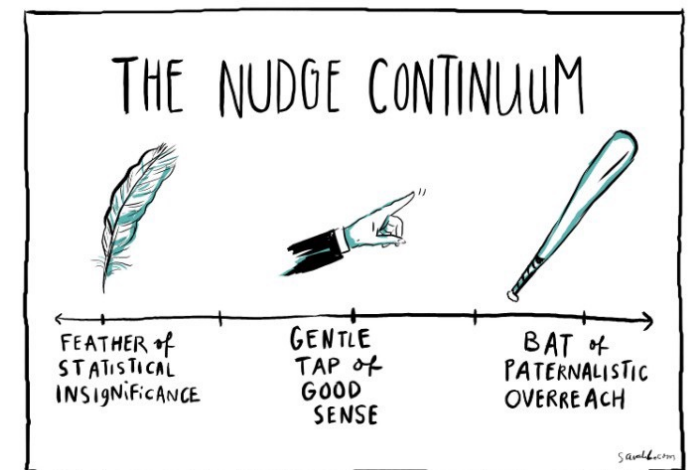


Solutions to social dilemmas

Presumed (opt-out) versus explicit (opt-in) consent in organ donorship

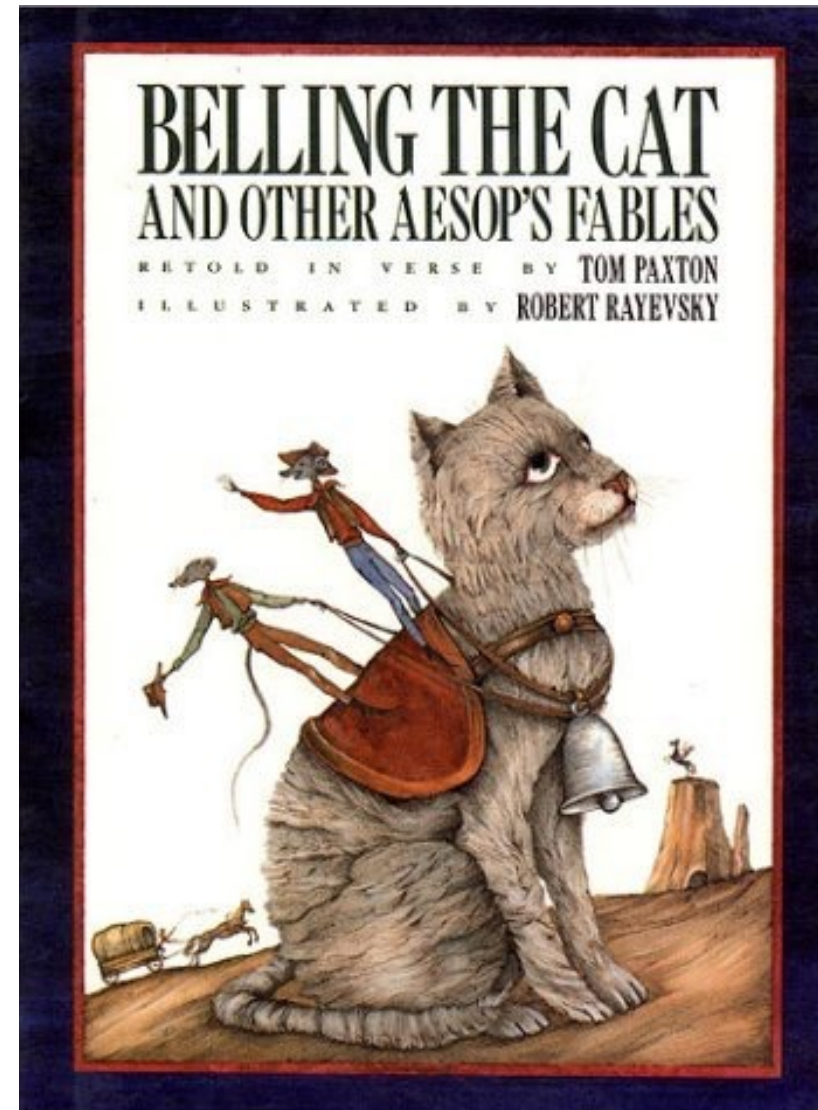


Effective consent rates, by country. Explicit consent (opt-in, gold) and presumed consent (opt-out, blue).



Social dilemmas and game theory

- Devising policies to promote people's wellbeing requires an understanding of the difference between:
 - situations in which self-interest can promote *general wellbeing*, and
 - situations in which self-interest leads to *undesirable results*
- In this unit we use the tools of [Game Theory?](#) — a way of modelling how people interact — to model social interactions and explain social dilemmas.



Social interactions and game theory

- On which side of road do you drive?
 - Government sets the rule
- If you have a free choice to choose on which side to drive
 - If the majority drives left, you will also choose to drive left
 - What is best for you in this case is also best for everybody else involved
 - However, what is best for you is not necessarily always the best for the community
- **Social interactions:** Two or more people interact, and their actions affect both their own outcome as well as other people's outcome
- We use game theory to modulate these interaction

Game theory: key concepts

- **Games** are models of **players'** strategic interactions
- **Strategic interaction** = when, in a social interaction involving two or more people (players), people are aware of how their actions affect both their own and other's outcomes, and vice versa
- **Strategy** = a choice that a player may take in a social interaction (the player is aware of the mutual dependence of that action's outcome on themselves and others)
- The **outcome** is the situation that results from the combination of strategies by both players
- A game states the **payoffs** earned by each player for each possible outcome / combination of strategies



Example: The “Invisible Hand” Game

See: [Invisible hand game](#)

Players – who is involved in the interaction

- Bala and Anil (farmers)

Feasible strategies – actions each player can take

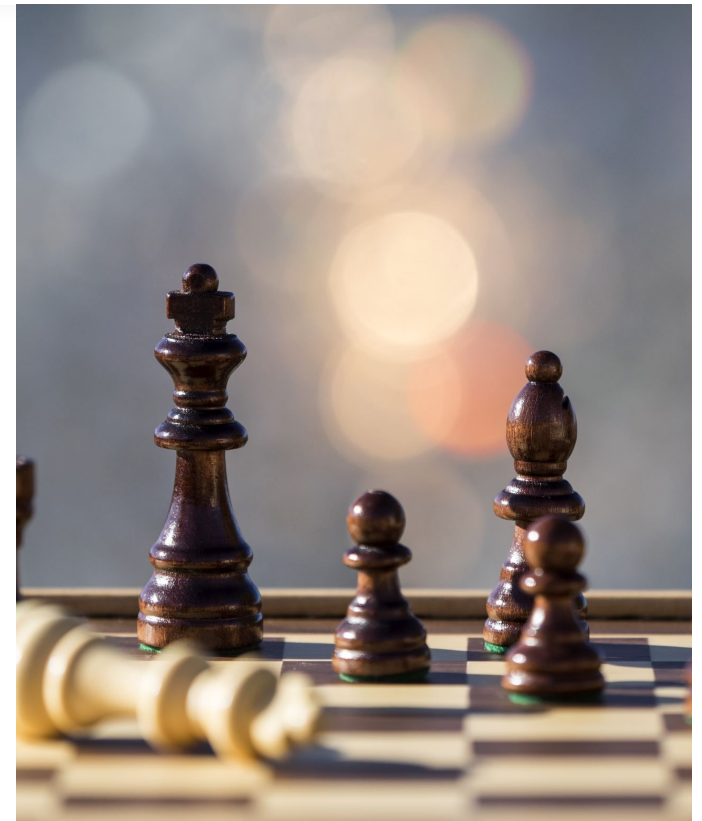
- Grow rice or cassava

Information – what each player knows when making their decision

- Anil's land is better suited to grow cassava
- Bala's land is better suited to grow rice
- The price they receive at market will be higher the less they bring

Payoffs – outcomes for every possible combination of actions

2 strategies × 2 players = 4 hypothetical outcomes



Assumptions

- There are no other people involved in this social interaction
- The only decision Anil and Bala need to make is which crop to grow
- Anil and Bala will interact just once ('one-shot' game)
- Their decisions are made simultaneously; they are both unaware of the decision that the other player made



Payoffs

- The payoffs represent the utility each player derives from each outcome.
- **Anil's land is equally suited for both crops, but Bala's land is more suited for rice**
- In this game there are 4 outcomes.
 - Each player has a preference ranking over the outcomes
 - The utility the player receives from the outcomes reflect this
 - The actual utility number is therefore arbitrary – think back to Unit 3. It only indicates the preferences of the players and the value that the players place on an outcome.
- Representation of the payoffs: row player first, followed by the column player's payoffs



The rice-cassava game: social interactions

		Bala	
		Rice	Cassava
Anil	Rice	Both produce rice There is a surplus of rice (low price) There is a shortage of cassava Anil is not producing cassava, which he is better able to produce	No market surplus High prices for both crops Both farmer are producing the crop for which they are less suited
	Cassava	No market surplus High prices for both crops Both farmer are producing the crop for which they are better suited	Both produce cassava There is a surplus of cassava (low price) There is a shortage of rice Bala is not producing rice, which he is better able to produce

The rice-cassava game: payoffs

		Bala	
		Rice	Cassava
Anil	Rice	Row player's payoff Column player's payoff 1, 3	2, 2
	Cassava	4, 4	3, 1

The textbook format

		Bala	
		Rice	Cassava
Anil	Rice	1 3	2 2
	Cassava	4 4	1 3

Equilibrium in the rice-cassava game

- **Best response** = strategy that will give a player the highest payoff, given the strategies the other players select (best response to an opponent's strategy)
- **Dominant strategy** = strategy that will give a player the highest payoff, *irrespective* what the other players do



The rice-cassava game: unique payoff values

4.2

- What is Anil's *best response* if:
 - Bala plants rice?
 - Bala plants cassava?
- For Anil, playing Cassava is a dominant strategy
- What is Bala's *best response* if:
 - Anil plants rice?
 - Anil plants cassava?
- For Bala, playing Rice is a dominant strategy.
- **Dominant strategy equilibrium** = an outcome of a game in which every player plays their dominant strategy

		Bala	
		Rice	Cassava
Anil	Rice	1, 3	2, 2
	Cassava	4, 4	3, 1

Fig 4.3 Dominant strategy equilibrium

- What is Anil's *best response* if:
 - Bala plants rice?
 - Bala plants cassava?
- For Anil, playing Cassava is a dominant strategy
- What is Bala's *best response* if:
 - Anil plants rice?
 - Anil plants cassava?
- For Bala, playing Rice is a dominant strategy.
- **Dominant strategy equilibrium** = an outcome of a game in which every player plays their dominant strategy

		Bala	
		Rice	Cassava
Anil	Rice	3, 5	4, 4
	Cassava	6, 6	5, 3

John Nash (1928-2015)

Nash's doctoral dissertation applied game theory to economics and introduced the idea of a ***Nash equilibrium***:

- At a Nash equilibrium, **no player has an incentive to unilaterally change their strategy.**
- The dominant strategy equilibrium in the invisible hand game is a type of Nash equilibrium.
- Nash and two colleagues won the Nobel prize in economics in 1994 “for their pioneering analysis of equilibria in the theory of non-cooperative games.”



Fig 4.2b Nash equilibrium

- What is Anil's *best response* if:
 - Bala plants rice?
 - Bala plants cassava?
- Anil does not have a dominant strategy
- What is Bala's *best response* if:
 - Anil plants rice?
 - Anil plants cassava?
- For Bala, playing Rice is a dominant strategy.
- Bala will play his dominant strategy of Rice, and if he plays rice it is best for Anil to play Cassava

		Bala	
		Rice	Cassava
Anil	Rice	4, 4	6, 3
	Cassava	6, 6	5, 2

Nash Equilibrium

Why is the outcome (Cassava, Rice) a Nash equilibrium?

		Bala	
		Rice	Cassava
Anil	Rice	4, 4	6, 3
	Cassava	6, 6	5, 2

The prisoners' dilemma: [Accuse vs Deny](#)

- **Two** prisoner's (Thelma & Louise)
- **Two** strategies:
 - Implicate the other prisoner in a crime (*accuse*)
 - Deny that the other prisoner was involved (*deny*)
- **Four** hypothetical outcomes
 - Both accuse, and receive 5 years each
 - Both deny, and receive 1 year each
 - Accuse whilst the other denies, with the accuser being freed for cooperation and the denier receiving 10 years



The prisoners' dilemma

		Louise	
		Deny	Accuse
Thelma	Deny	1, 1	10, 0
	Accuse	0, 10	5, 5

Note: Payoffs are in years of imprisonment so players prefer lower numbers (disutility)

What is the equilibrium outcome of this game?

Is it a Nash equilibrium?

The prisoners' dilemma

		Louise	
		Deny	Accuse
Thelma	Deny	1, 1	10, 0
	Accuse	0, 10	5, 5

What is the equilibrium outcome of this game?

Is it a Nash equilibrium?

The prisoners' dilemma



- Bala and Anil now face a different problem:
- → *How to deal with pests?*
- **Feasible strategies:**
 - Use an inexpensive chemical (*Toxic Tide*) that kills all insects, but contaminates the water that they both drink
 - Use an integrated pest control (*IPC*) that introduces beneficial insects as a natural predator for pests
- **Information:**
 - Their fields are adjacent
 - If one player chooses Toxic Tider the damage is limited, but if both choose Toxic Tide the damage is severe

Fig 4.4a The pest control game: social interactions

		Bala	
		IPC	Toxic Tide
Anil	IPC	Beneficial insects spread over both fields, eliminating pests No water contamination	Bala's chemicals spread to Anil's field and kill his beneficial insects Limited water contamination
	Toxic Tide	Anil's chemicals spread to Bala's field and kill his beneficial insects Limited water contamination	Eliminates all pests Heavy water contamination Requires costly filtration system

Fig 4.4b Pest control game

		Bala	
		IPC	Toxic Tide
Anil	IPC	3, 3	1, 4
	Toxic Tide	4, 1	2, 2

If Bala chooses IPC

Anil chooses Toxic Tide

If Bala Chooses Toxic Tide

Anil chooses Toxic Tide

Anil's dominant strategy is Toxic Tide

If Anil chooses IPC

Bala chooses Toxic Tide

If Anil Chooses Toxic Tide

Bala chooses Toxic Tide

Bala's dominant strategy is Toxic Tide

Outcome of game is Toxic Tide/Toxic Tide

Both receives a payoff of 2, but both would have been better off if they both used IPC (3 each)

Leads to a sub-optimal outcome

This is called the prisoners' dilemma

The prisoners' dilemma

The outcome (Toxic Tide, Toxic Tide) is a *Nash equilibrium* **AND** a *dominant strategy equilibrium*.

Both players would be better off with the outcome (IPC, IPC).

		Bala	
		IPC	Toxic Tide
Anil	IPC	3, 3	1, 4
	Toxic Tide	4, 1	2, 2

Where do
we see
prisoners'
dilemmas?



Traffic jams



Climate change

Pareto criterium

- We call the outcome of an economic interaction an **allocation**.
- Suppose that we want to compare two possible allocations, A and B, that may result from an economic interaction.
- According to the **Pareto criterion**, allocation A is better than allocation B if at least one party would be strictly better off with A than B, and nobody would be worse off.
- We say that A **Pareto dominates** B, or that A would be a **Pareto improvement** over B.

Evaluating efficiency

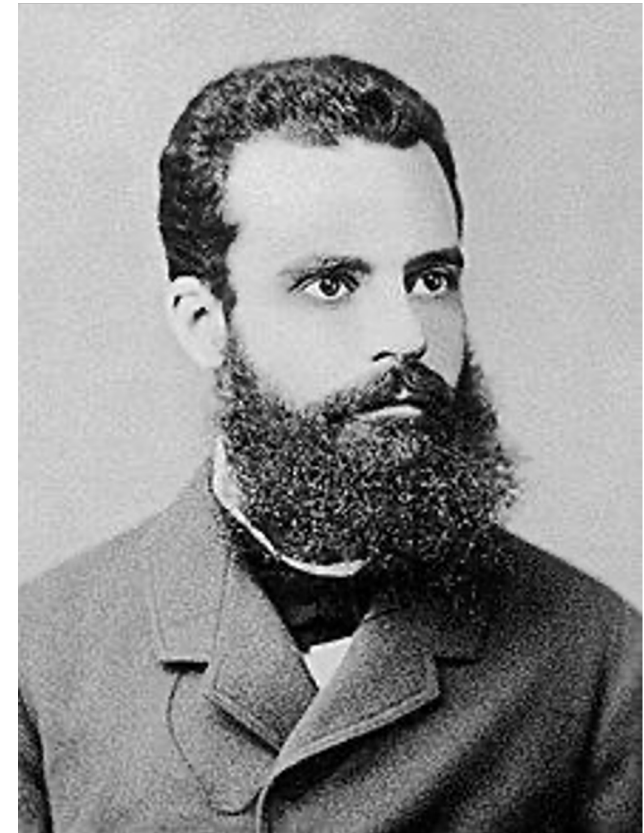
- In economics, efficiency does *not* mean:
 - the most sensible way of doing something (engineering)
 - the most profitable way of doing something (business)
- An allocation is ***Pareto efficient*** if there is no other allocation that makes at least one person better off and no person worse off.
- If an allocation is not efficient, it is *inefficient*.



Is poverty efficient?

Vilfredo Pareto (1848-1923)

- Pareto earned a degree in engineering and sought to make economics and sociology fact-based sciences similar to the physical sciences.
- The *Pareto law* holds that a small group of people always own most wealth (80-20 in Pareto's time, 90-20 in the U.S. today).
- Given this inequality, Pareto argued that people fight over their share of the pie:
"[T]he efforts of men are utilized in two different ways: they are directed to the production or transformation of economic goods, or else to the appropriation of goods produced by others." Vilfredo Pareto (1906)
- Pareto is most famous today for defining the concept of efficiency in economics.

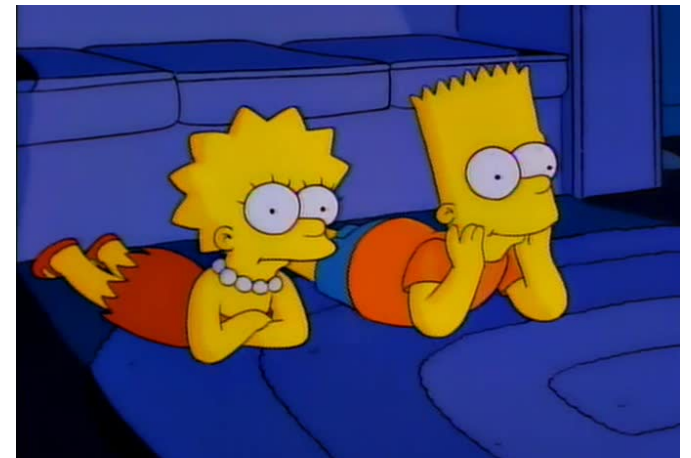


Pareto efficiency vs Pareto inefficiency

- A **Pareto improvement** is said to occur when a change in allocation **makes at least one person better off and nobody else worse off**, given an initial allocation of goods for a set of persons.
 - In other words, there was no tradeoff involved in changing from an initial allocation to an alternative allocation
- The initial allocation is then said to be **Pareto inefficient**
- An allocation is **Pareto efficient** if there are no alternative allocations that would lead to a Pareto improvement
 - A change to an alternative allocation implies a tradeoff: no one person is made better off without making at least one other person worse off

Pareto efficiency vs Pareto inefficiency

- Consider a household with only one television.
- A brother and sister equally prefer spending time watching TV, but they do not enjoy watching the same TV shows.
- If they are allowed to watch 4 hours of TV a day, how can this good (TV time) be efficiently allocated between them?
- For example, is an allocation of 2 hours of TV time each Pareto efficient?
- To answer this, we need to determine if other alternative allocations of TV time allow for Pareto improvements.



Pareto efficiency vs Pareto inefficiency

- If instead of allocating 2 hours of TV each, we allocated 3 hours to the brother and 1 hour to the sister, the brother would be better off and the sister would be worse off.
- This allocation does not represent a Pareto improvement of the allocation 2 hours each
- What if we gave the sister 2.5 hours of TV time and the brother 1.5 hours?
- Again, compared with 2 hours each, the sister is better off and the brother is worse off.

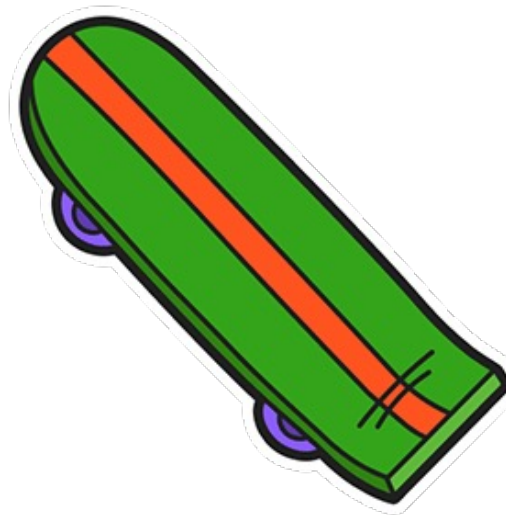
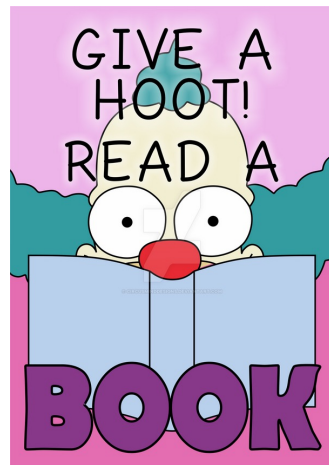


In our example, there is no allocation of TV time that would make either sibling better off without making someone worse off.

As long as all resources (4 hours of TV time) are being allocated, Pareto improvements are impossible.

All allocations of the full 4 hours of TV between the siblings are Pareto efficient.

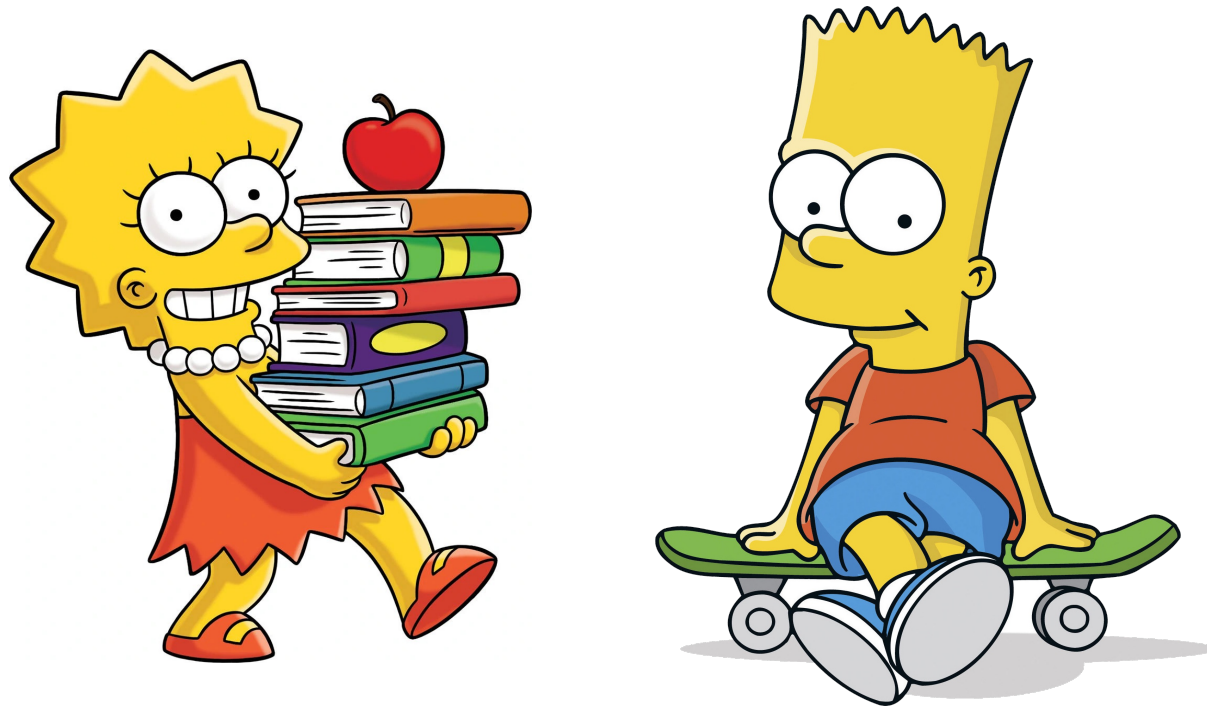
- Consider the same two siblings, but now there are two items: a book and a skateboard
- One of the siblings (Bart) prefers books to skateboards, and the other (Lisa) prefers skateboards to books.
- Are there any Pareto efficient allocations of these two goods?



- Let's suppose their parents got them what they prefer for Christmas, but they mislabeled the presents
- Lisa ended up receiving a skateboard, and Bart a book
- Is this allocation Pareto efficient?



- There is **one Pareto efficient allocation** of these two items.
- Lisa(who prefers books) gets the book, and Bart (who prefers skateboards) gets the skateboard.



Mutually beneficial exchange and Pareto efficiency

After children return from trick-or-treating on Halloween, they often engage in trading candy until all mutually beneficial trades are exhausted.

Every trade that is made improves one child's utility without hurting another child's utility.

The children eventually succeed in reaching a Pareto-efficient distribution of candy.

Any exchange that is mutually beneficial signals that the original allocation was Pareto inefficient.

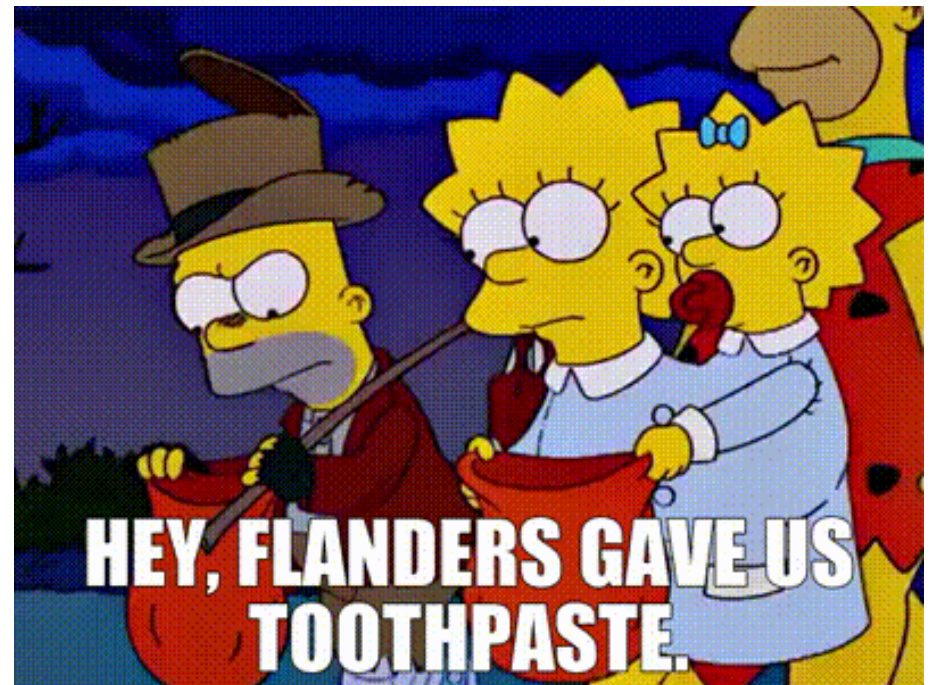
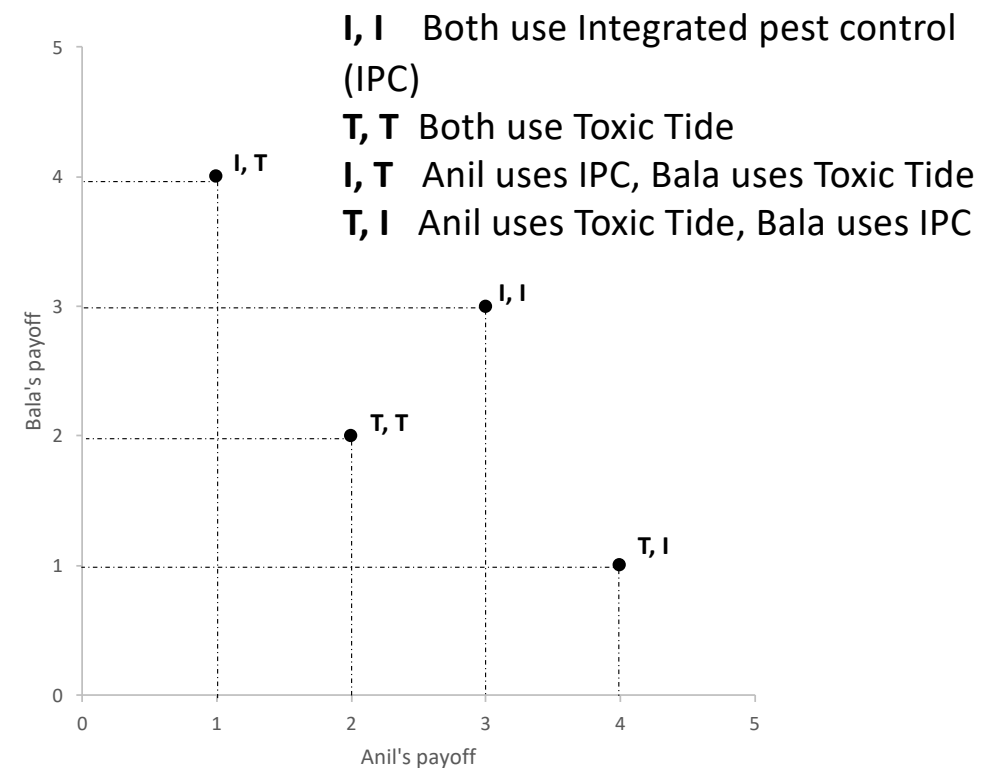


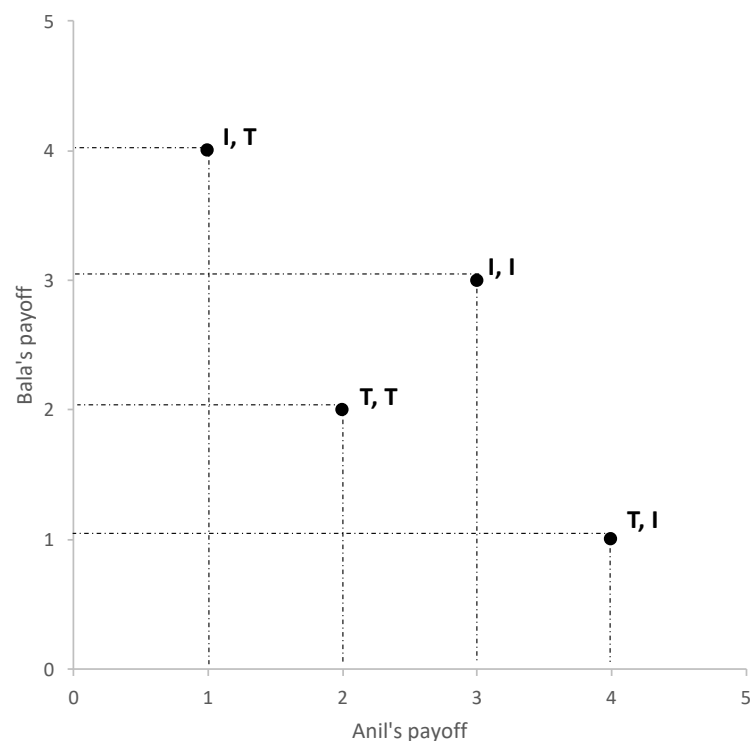
Fig 4.6 The Pareto criterion

		Bala	
		IPC	Toxic Tide
Anil	IPC	3 3	4 1
	Toxic Tide	1 4	2 2



The right-hand graph illustrates the payoff matrix of the pesticide example

Fig 4.6 The Pareto criterion



Allocation at (I, I) Pareto dominates allocation (T, T) – both players will be better at (I, I) than at (T, T)

Shift from allocation (T, T) to (I, I) will be better for Anil, but Bala will be worse off, and a shift from allocation (T, T) to (I, T) will be better for Bala, but Anil will be worse off

No allocation lies north-east of allocation (I, I) – so it is not Pareto dominated.

No allocation lies north-east of allocation (I, T) or allocation (T, I) – so it is not Pareto dominated, but also not Pareto dominate any other allocation

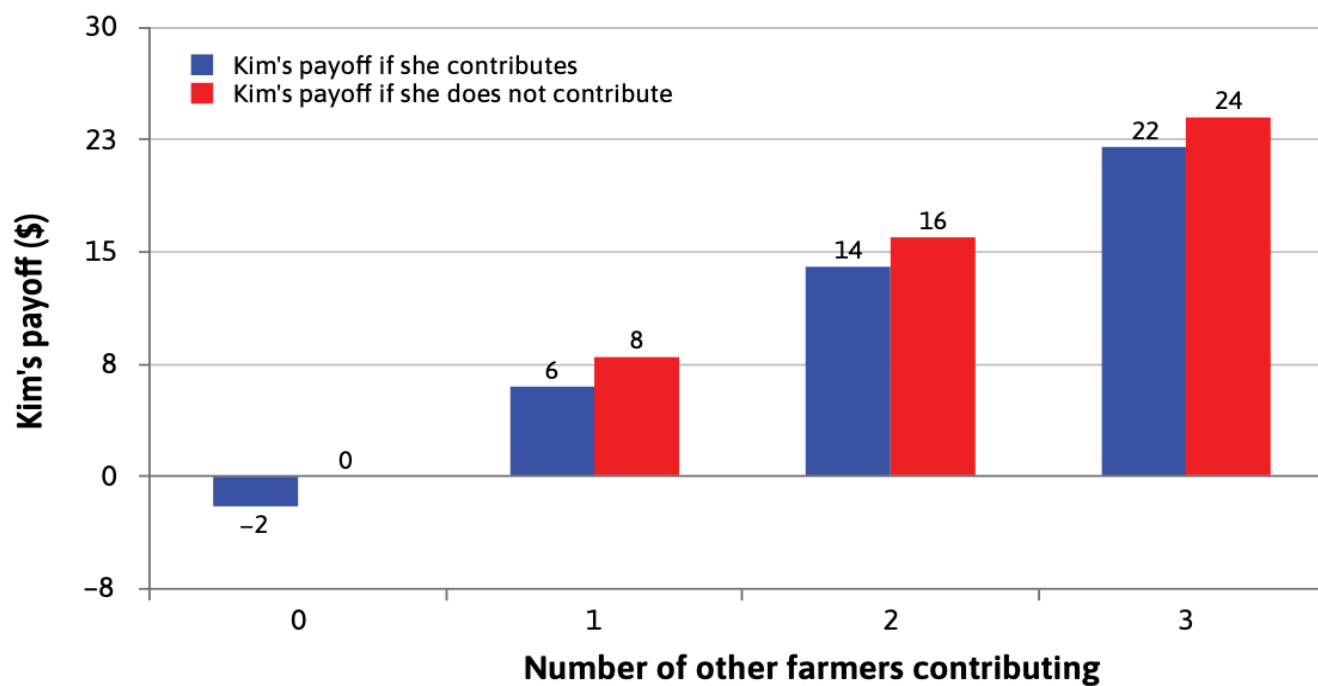
Pareto criterium

- No value judgement is made in declaring that a Pareto allocation is where you cannot increase the welfare of one person while not simultaneously worsen the welfare of somebody else.
- If an allocation is not Pareto-dominated by any other allocation it is Pareto-efficient.
- In the previous example combinations (I, T), (I, I) and (T, I) are Pareto-efficient (so it is NOT unique) and allocations (T, T) is not Pareto efficient, because it is dominated by allocation (I, I).
- Pareto criterium does not tell us which Pareto-efficient allocation is best
- A Pareto-efficient allocation is not always better than a Pareto-inefficient one. (T, T) is Pareto-efficient and (T, T) not, but (T, T) does not dominate (T, T).
- We may want to assess both the fairness and Pareto-efficiency when evaluating outcomes

Public goods game

- Four farmers must decide on their contribution to an irrigation project
- Cost of contributing for each farmer is \$10
- Every contribution of \$10 increases the crop yield on each of the four farms by \$8

Fig 4.9 Kim's pay-offs in irrigation game



Kim's pay-off is the highest if she does not contribute (**red bars**) than when she does (**blue bars**)

Public good: When one individual bears a cost to provide a good everybody benefits

In a public goods game, everybody can **free-ride** on the contributions of others.

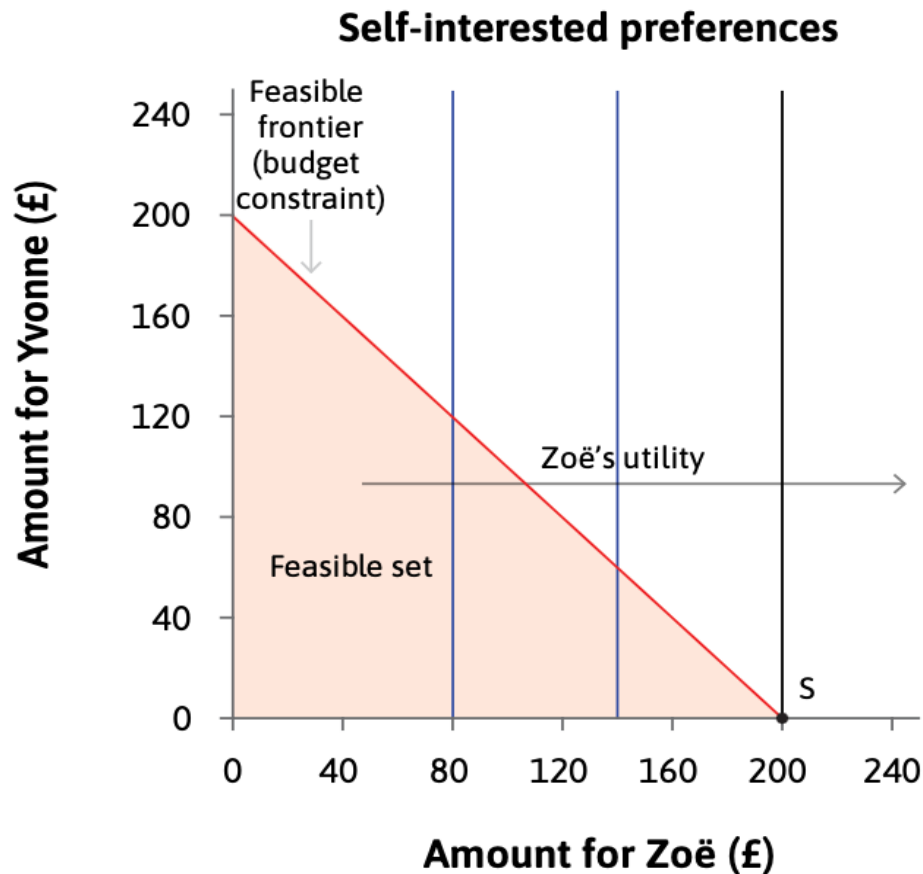
Irrigation game

- The irrigation game is a prisoners' dilemma with more than two players
- If farmers only care about their own monetary pay-off the dominant strategy is where no one contributes, and the pay-offs are zero.
- If everybody contributed each would get \$22
- Every farmer would benefit if everyone cooperated
- Cooperation cannot be an equilibrium, because each farmer can do better by free-riding the others

Altruism

- Sometimes people has social preferences that their utility depends not only on what they obtain themselves, but also by the wellbeing of others
- Altruism: A social preference in which an individual's utility is increased by benefits to others.
- Suppose Zoë enters the lottery and wins \$200. Does she decide to keep the \$200 or does she share it with her flatmate Yvonne?
- This is a model where Zoë must decide between two “goods”: Her own share and if she is altruistic, Yvonne's share

Fig 4.12 Outcome if Zoë is self-interested

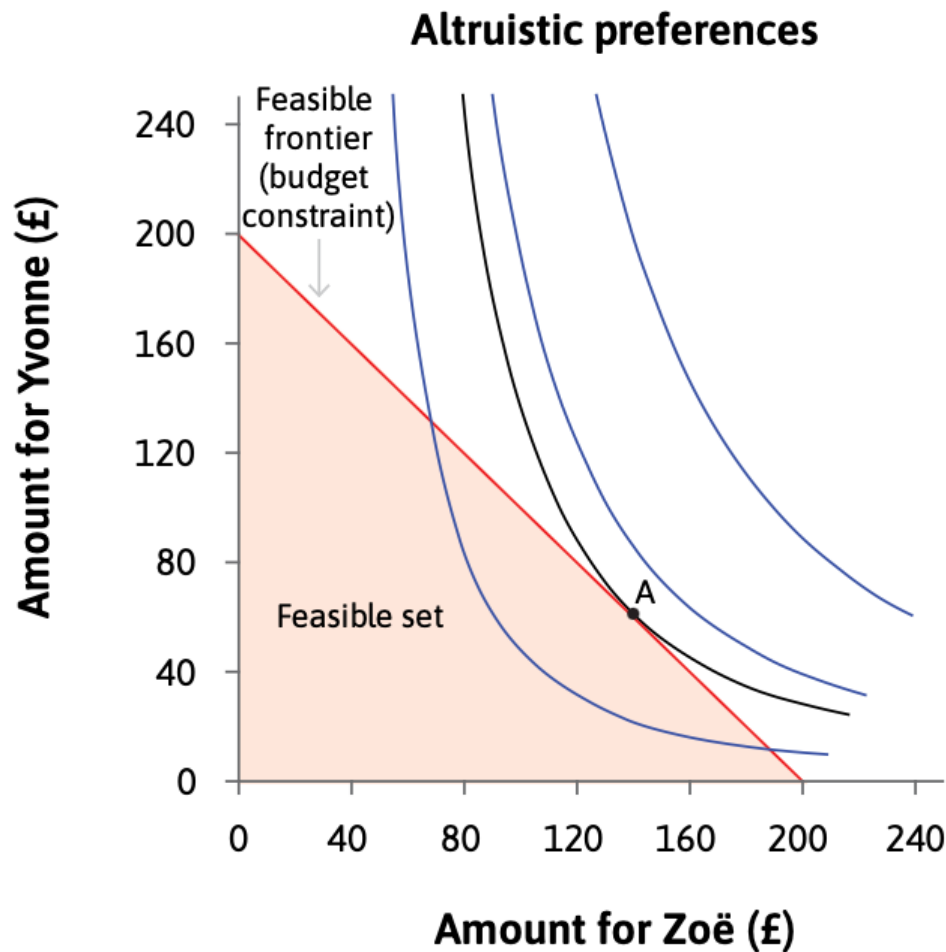


Feasible frontier is straight line that intersects *Amount for Zoë* axis at \$200 if she is self-centered and the *Amount for Yvonne* axis at \$200 if she is totally altruistic

If Zoë is self-centered she cares only about herself, and her indifference curves will be vertical lines

Her utility would be highest at Point S where she keep the \$200 and gives Yvonne nothing

Fig 4.11 Outcome if Zoë is altruistic



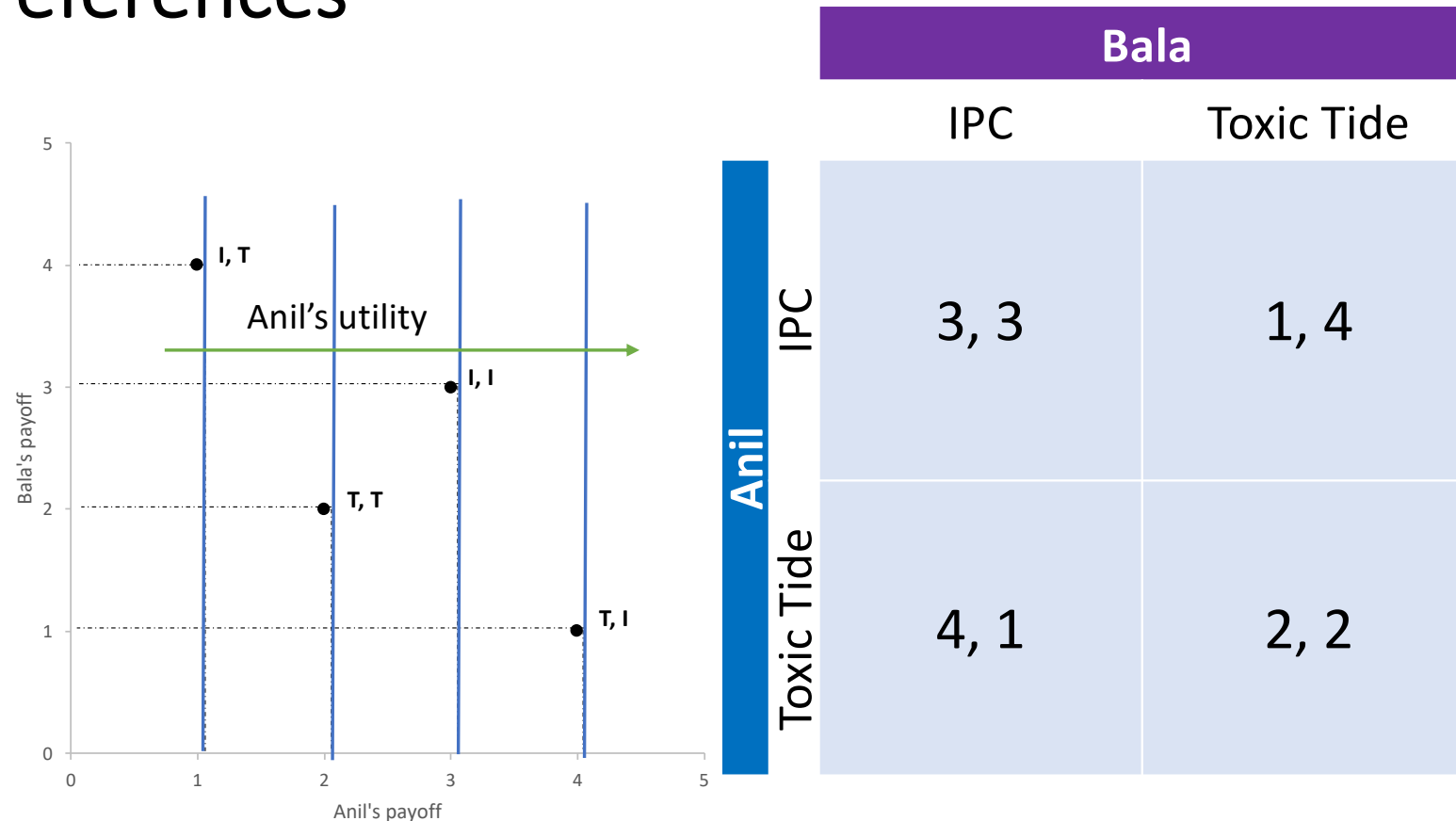
If Zoë is altruistic her utility will also be based on what Yvonne will receive

Therefore her indifference curves are the normal convex shaped curves

She will maximise her utility at A where her indifference curve is tangent to the budget constraint

She will keep \$140 and give \$60 to Yvonne

Fig 4.12 Prisoners' dilemma with self-interested preferences

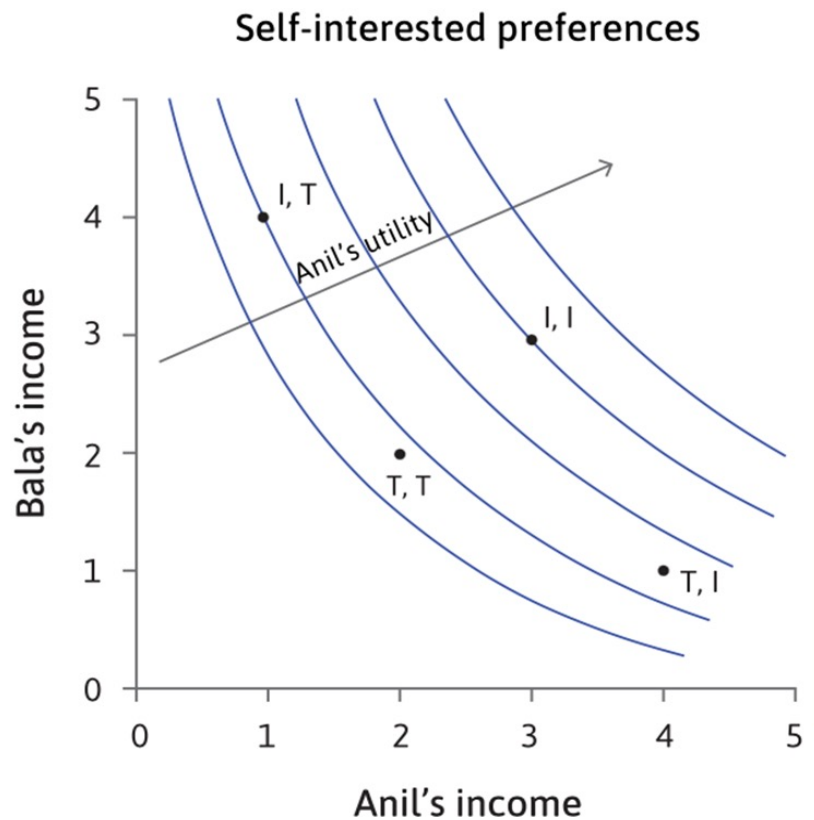


If Bala Chooses Toxic Tide Anil will also choose Toxic Tide

If Bala Chooses IPS Anil will again choose Toxic Tide

Toxic Tide is her dominant strategy

Fig 4.13 Prisoners' dilemma with altruism



		Bala	
		IPC	Toxic Tide
Anil	IPC	3, 3	1, 4
	Toxic Tide	4, 1	2, 2

Anil cares about Bala's welfare. Therefore his indifference curves are downward sloping

If Bala chooses Toxic Tide Anil will choose IPC, because (I, T) is on higher indifference curve than (T, T)

If Bala chooses IPC Anil will also choose IPC because (I, I) is on higher indifference curve than (T, I). (I, I) gives him lower income, he prefers it as it does not inflict damage on Bala

Learning about preferences

Economists sometimes use experiments to learn about preferences.

1. Lab experiments:

- Can control participants' decisions and their outcomes.
- Can create a control/treatment group for comparison.
- Results can be replicated.
- Can control for other variables.

2. Field experiments:

- Lab experiments may not predict real-world decision making.
- More realistic context in which people make decisions.



Ultimatum game – one shot game

- The **ultimatum game** is a common experimental economics game in which two parties interact anonymously and only **once**
- The ultimatum game is used to study preferences
- It has been used around the world with experimental subjects including students, farmers, warehouse workers, and hunter gatherers
- By observing their choices, we investigate the subjects' preferences and motives, such as pure self-interest, altruism, inequality aversion, or reciprocity



Ultimatum game – one shot game

- Random pairs of players (identities held anonymous) are created, with one randomly assigned in the role of “Proposer” and the other in the role of “Responder”
- Proposer is given an amount of money, say R100, and instructed to offer the Responder part of it

“ x for me, y for you”

- After observing the offer, the Responder *can either accept or reject it* (take-it-or-leave-it offer)
- If rejected, both get nothing; if accepted, the Proposer gets x and the Respondent gets y



Ultimatum game – one shot game

- Let's say the Proposer offers R35 and the Responder accepts
→ the Proposer gets R65 and the Responder gets R35
- If the Responder rejects the offer
→ they both get nothing.
- This is called a take-it-or-leave-it offer. It is *the ultimatum* in the game's name.
- The Responder is faced with a choice: accept R35 or reject and get nothing.
→ Therefore, R35 is the opportunity cost of rejecting the offer
- The game is about sharing economic rents that arise in an interaction



Ultimatum game – one shot game

- We see that in the ultimatum game one player (the Proposer) chooses her strategy first, *followed* by the Responder.
 - This is called a **sequential game**; previously we looked at **simultaneous games**, in which players chose strategies simultaneously.
- What the Proposer will get depends on what the Responder does, so the Proposer has to think about the likely response of the other player.
- That is why this is called a **strategic interaction**.

What should the Proposer do?



Figure 4.16. Game tree for a simplified ultimatum game.

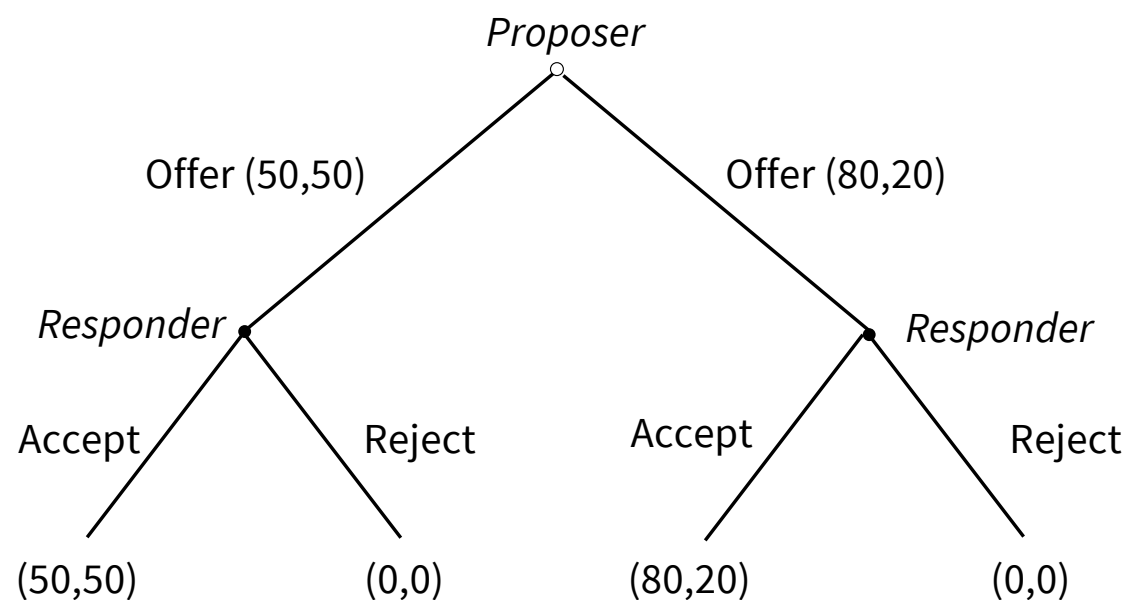
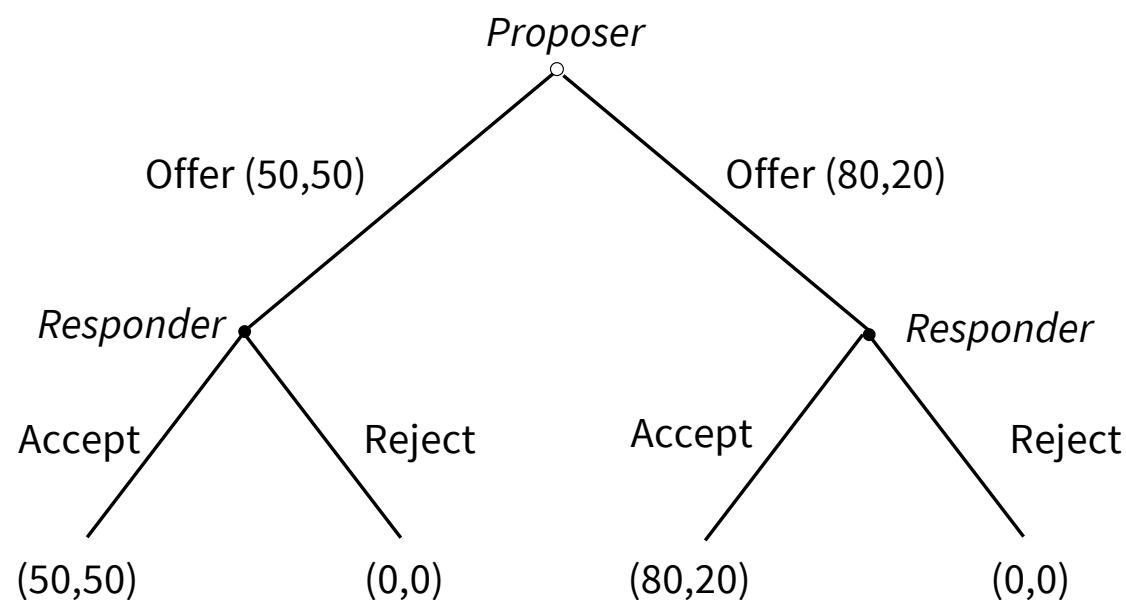
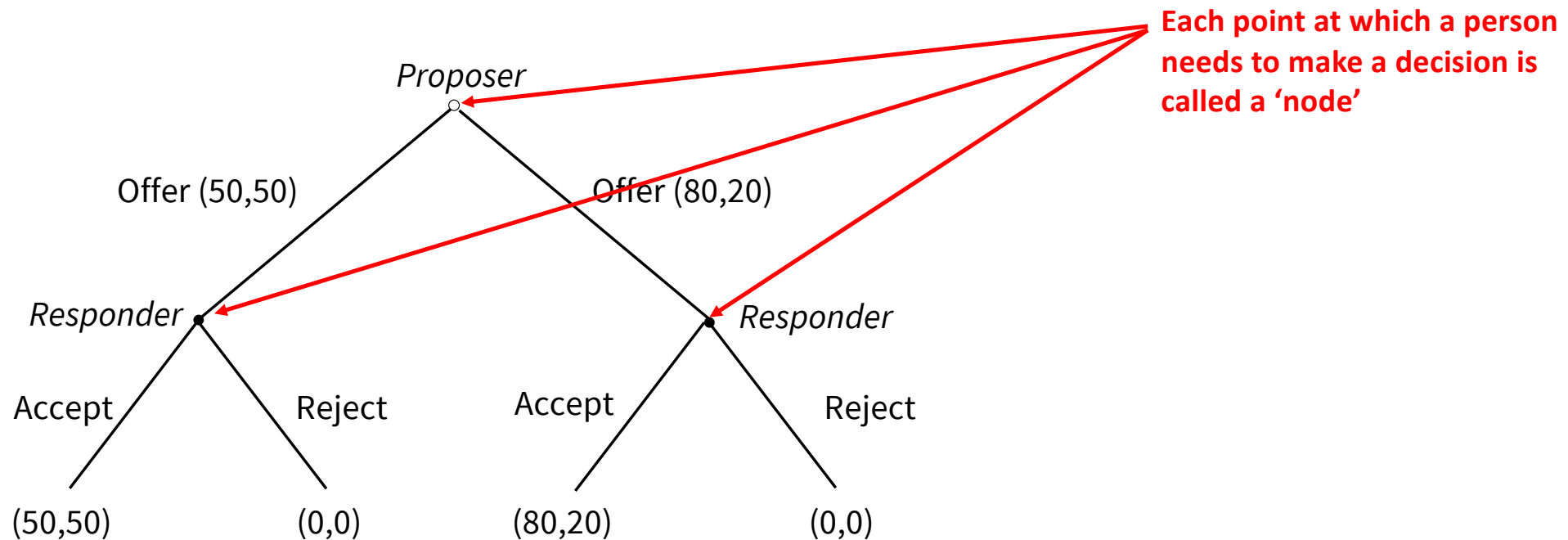


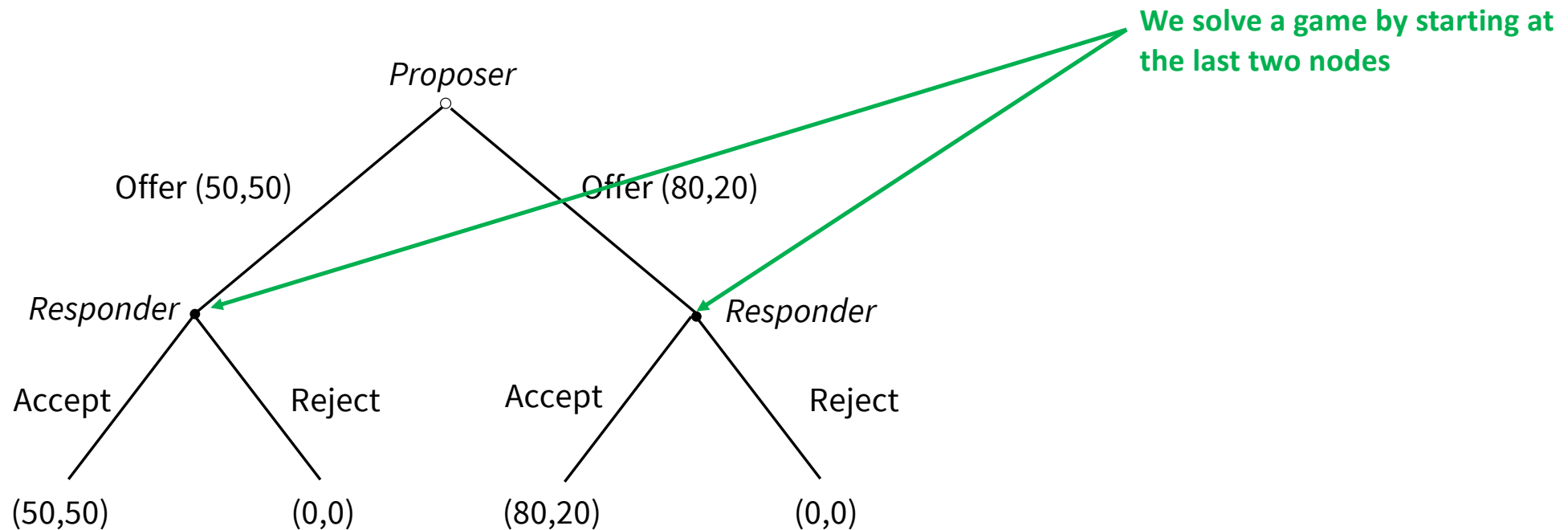
Figure 4.16. Game tree for a simplified ultimatum game.



How do we solve a game tree for a simplified ultimatum game?



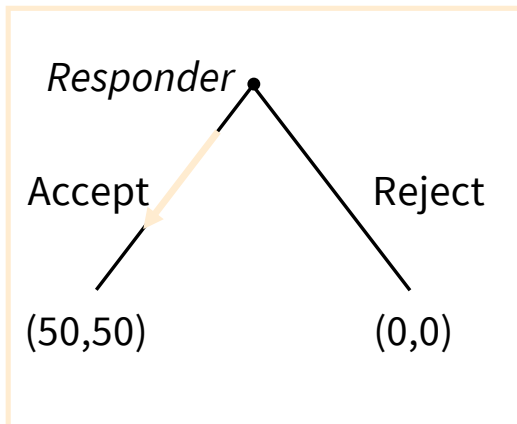
How do we solve a game tree for a simplified ultimatum game?



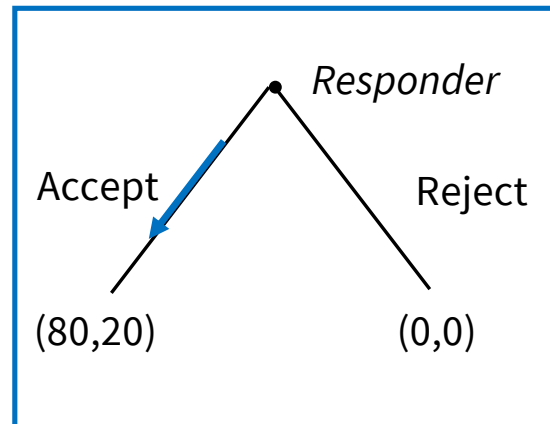
How do we solve a game tree for a simplified ultimatum game?

If you were the responder what would you choose?

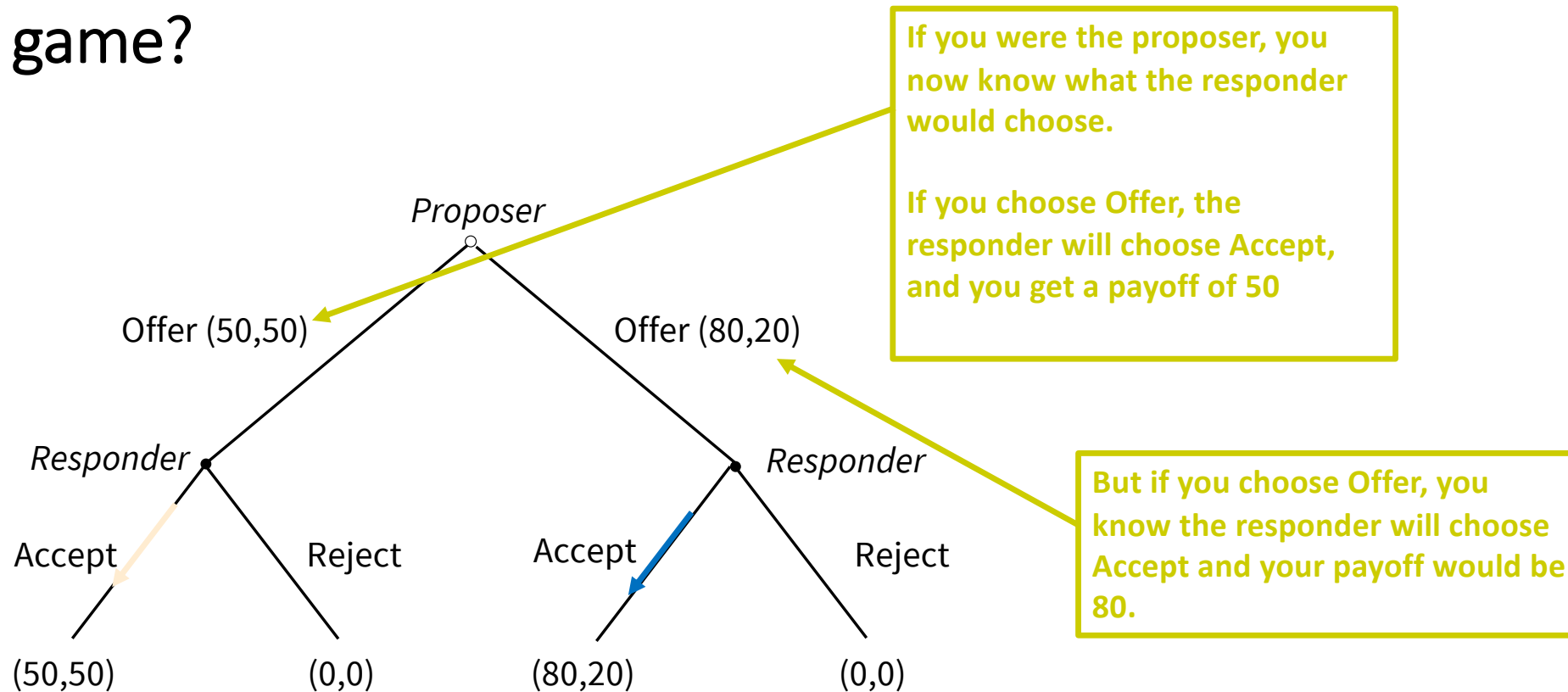
Accept and get 50



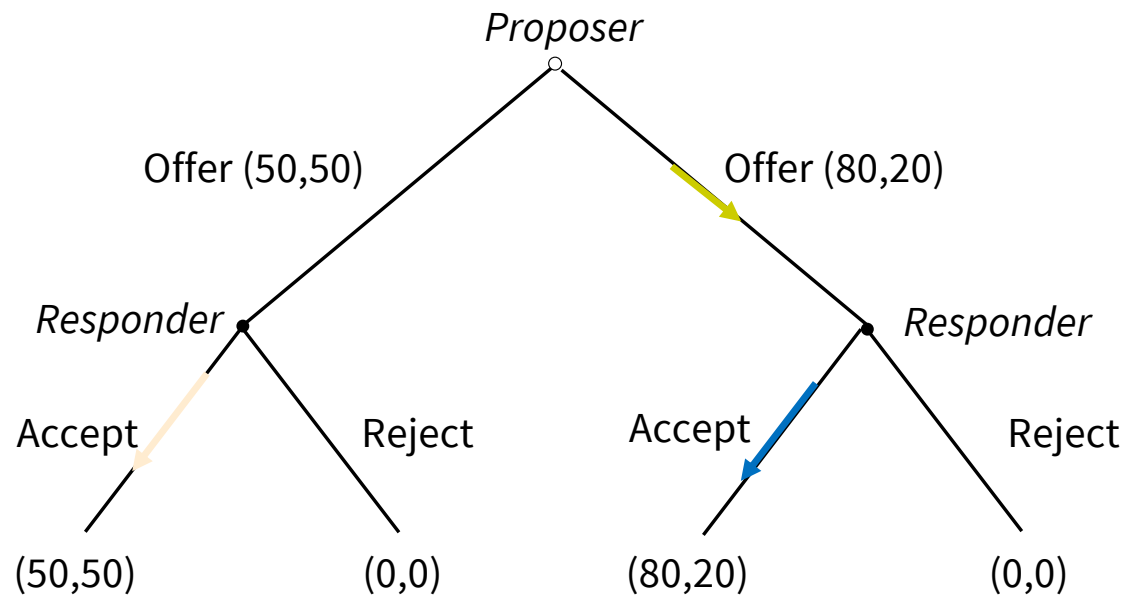
Accept and get 20



How do we solve a game tree for a simplified ultimatum game?



How do we solve a game tree for a simplified ultimatum game?



And the full solution to the game is:
(Offer; Accept, Accept)

Ultimatum game – one shot game



Outcome of the game:



The Responder will accept any offer as it places them in a better position (getting something is better than getting nothing)



The Proposer should anticipate this and offer the lowest possible offer to receive the highest possible payoff

The Proposer offers R1

The Responder accepts this R1 since they are better off than they are when rejecting the offer (get R0)

Proposer thus gets R99



In real word → much higher splits, usually around 50-50. The actual split differs from the predicted split due to social preferences.

Class exercise:

If two individuals have to divide 10 pounds between themselves using the ultimatum game:

- a. If they were both selfish, and assuming the Responder accepts offers if he is indifferent between accepting and not accepting, they would be willing to pay up to 10 pounds to be able to be the proposer of the game.
- b. Without information about the Responder's preferences, completely selfish Proposers may still make positive offers.
- c. The responder would prefer not to have to compete with other responders in the game.
- d. All of the above

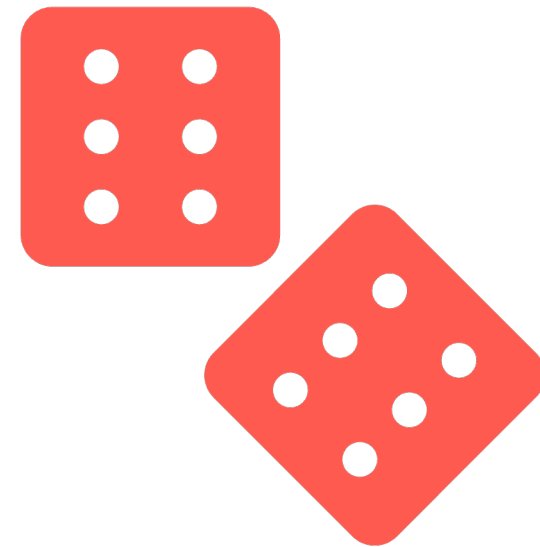


Fig 4.17 Fair farmers or self-interested students?

- An experiment using the ultimatum game was run with a group of farmers in Kenya, and a group of students in the US. Proposers could offer 0, 10, 20, 30, 40, or 50% of the pie to Responders

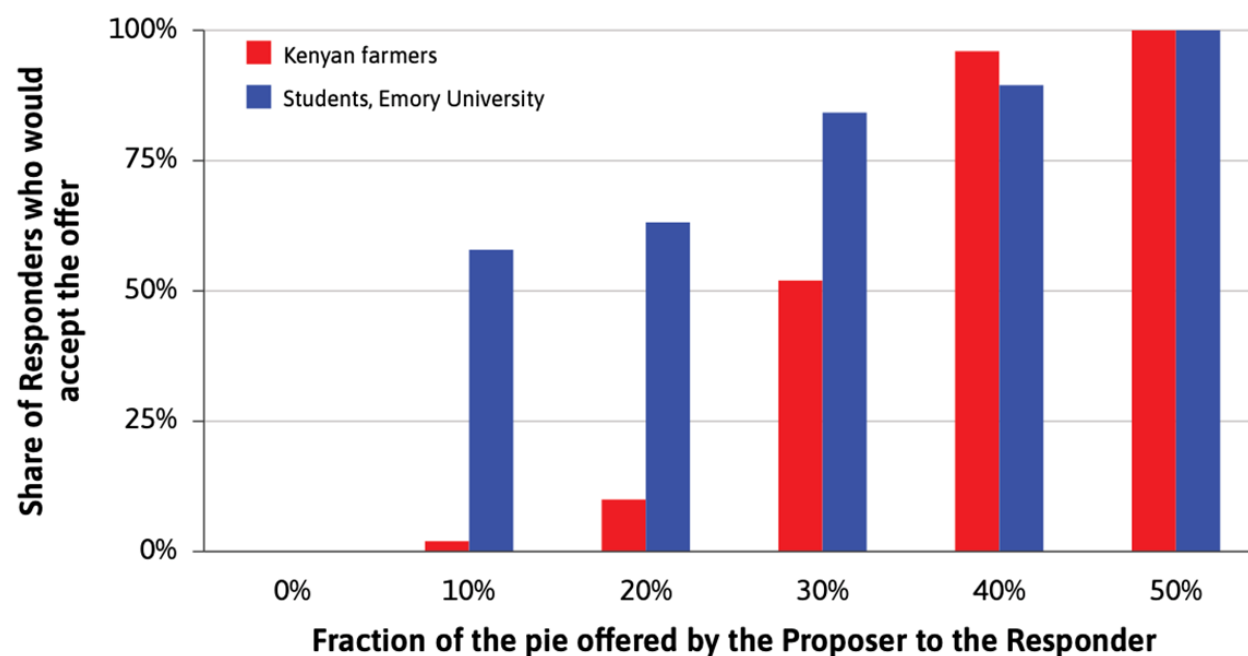


Fig 4.21 Game with two Nash Equilibria

		Bala	
		Rice	Cassava
Anil	Rice	2, 3	4, 4
	Cassava	6, 6	3, 2

If Bala chooses Rice, Anil will choose Cassava

If Bala chooses Cassava, Anil will choose Rice

Anil does not have a dominant strategy

If Anil chooses Rice, Bala chooses Cassava

If Anil chooses Cassava, Bala chooses Rice

Bala does not have a dominant strategy

Anil and Bala prefer to do the opposite whatever their neighbour does

(Cassava, Rice) and (Rice, Cassava) are Nash equilibria

This is a **coordination game**

Coordination game with conflict of interest

- Conflict of interest in a coordination game is when the players in the game prefer different Nash equilibria
- Astrid and Bettina are two Software engineers working on a project
 - Decide to write code in Java or C++
 - They each have to choose one language
 - Astrid wants to write in Java because she is better at writing Java code
 - Bettina prefers C++ because she is better at writing in C++ code

Fig 4.22 Pay-offs in a programming game with a conflict of interest

		Bettina	
		Java	C++
Astrid	Java	4, 3	2, 2
	C++	0, 0	3, 6

Both do better if they write in one language

Astrid does better if that language is Java

Bettina does better if that language is C++

Two Nash Equilibria: Both choose Java or both choose C++

Their total pay-off is the highest if they both choose C++

Negotiations are needed to resolve the conflict of interest

They could agree to both use C++ and split the proceeds in a way that will make both content with the outcome

In the next unit

- The role of **institutions** in social allocations
- How bargaining power affects the distribution of surplus

Class exercise:

Assume an economy comprises only two firms, Firm A and Firm B. The following table shows the profits (or losses if negative) when the firms invest or do not invest (the first entry in each box represents Firm A). Based on this information, which of the following statements is/are correct?

	Firm B invests	Firm B does not invest
Firm A Invests	400, 60	-100, 50
Firm A does not invest	310, -10	55, 45

- a. Investing is a dominant strategy for at least one firm.
- b. The Pareto efficient outcome is a Nash equilibrium.
- c. Firm B is more dependent on investment by Firm A than the converse.
- d. All of the above.